

Recursive Macroeconomic Theory

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Complete Solutions

Chapter 7 — Recursive Competitive Equilibrium: I

All 7 end-of-chapter exercises, 7.1–7.7 (pp. 244–248)

What is in this file

Exercises 7.1–7.5 are one industry seen five ways: a price-taking firm with arbitrary beliefs (7.1), the same firm with rational expectations (7.2), a planner (7.3), a monopolist (7.4) and a duopoly (7.5). All are linear regulators, solved with the program of Exercise 5.1(b) (`olrp`, Python instead of Matlab) and, for the duopoly, a Python `nnash`. Exercises 7.6 (time-inconsistent preferences) and 7.7 (equilibrium search) are conceptual; each gets a worked numerical illustration with parameters stated where it is used.

The companion notebook `ls_ch07.ipynb` (same folder) computes every number quoted here and checks each one by an independent route with an `assert`: Riccati iteration vs `scipy`'s DARE vs undetermined coefficients; the rational-expectations fixed point vs the planner; `nnash` vs best-response iteration; grid solutions vs closed forms.

Question boxes reproduce the full statement. Blue boxes are odd-numbered exercises, purple boxes even-numbered ones.

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Chapter 7 Recursive Competitive Equilibrium: I

Chapter 7 Overview

Big K , little k . A price-taking agent's state has two parts: x , which it controls, and X , the same variable chosen "by the market", which it only forecasts through a *perceived* law $X' = G(X)$ (7.3.3). Its Bellman equation (7.3.1) gives a rule $u = h(x, X)$ (7.3.5). Setting $x = X$ after optimising gives the *actual* law $X' = G_A(X) \equiv g[X, X, h(X, X)]$ (7.3.6).

Recursive competitive (rational expectations) equilibrium (p. 231): h, G_A, G such that h is optimal given G and $G_A = G$. In the adjustment-cost model of section 7.2 the map from perceived to actual laws is $H \mapsto \mathcal{M}(H)$, $H(Y) = nh(Y/n, Y)$ (7.2.10), and an equilibrium is a fixed point of $\mathcal{M}$. $\mathcal{M}$ need not be a contraction.

Planning shortcut (section 7.2.1). With $n = 1$, the planner who maximises $\sum \beta^t S_t$, $S_t = \int_0^{Y_t} (A_0 - A_1 x) dx - .5d(Y_{t+1} - Y_t)^2$ (7.2.11), has Euler equation (7.2.16), which is the firm's Euler equation (7.2.7) with $y_t = Y_t$. The planner's policy is the equilibrium law H .

Markov perfect equilibrium (sections 7.6–7.7). Without price taking, each player's Bellman equation takes the rival's *policy* as given (7.6.4)–(7.6.5). In linear-quadratic games this gives stacked Riccati equations (7.7.4)–(7.7.7), iterated backward to a stationary pair (F_1, F_2) (the book's `nnash.m`).

The regulator used throughout (chapter 5): maximise $-\sum_t \beta^t \{x'_t R x_t + u'_t Q u_t\}$ s.t. $x_{t+1} = A x_t + B u_t$. The solution is $v(x) = -x' P x$, $u = -F x$, with $P = R + \beta A' P A - \beta^2 A' P B (Q + \beta B' P B)^{-1} B' P A$ and $F = \beta (Q + \beta B' P B)^{-1} B' P A$.

Parameter values used in 7.1–7.5 (given in Exercise 7.1(c))

$A_0 = 100$, $A_1 = .05$, $\beta = .95$, $d = 10$; perceived law in 7.1: $H_0 = 95.5$, $H_1 = .95$.

Competitive Equilibrium with Adjustment Costs

Exercise 7.1 A competitive firm

Question

p. 244. A competitive firm seeks to maximize

$$\sum_{t=0}^{\infty} \beta^t R_t \quad (1)$$

where $\beta \in (0, 1)$, and time t revenue R_t is

$$R_t = p_t y_t - .5d(y_{t+1} - y_t)^2, \quad d > 0, \quad (2)$$

where p_t is the price of output, and y_t is the time t output of the firm. Here $.5d(y_{t+1} - y_t)^2$ measures the firm's cost of adjusting its rate of output. The firm starts with a given initial level of output y_0 . The price lies on the market demand curve

$$p_t = A_0 - A_1 Y_t, \quad A_0, A_1 > 0 \quad (3)$$

where Y_t is the market level of output, which the firm takes as exogenous, and which the firm believes follows the law of motion

$$Y_{t+1} = H_0 + H_1 Y_t, \quad (4)$$

with Y_0 as a fixed initial condition.

- a. Formulate the Bellman equation for the firm's problem.
- b. Formulate the firm's problem as a discounted optimal linear regulator problem, being careful to describe all of the objects needed. What is the *state* for the firm's problem?
- c. Use the Matlab program `olrp.m` to solve the firm's problem for the following parameter values: $A_0 = 100$, $A_1 = .05$, $\beta = .95$, $d = 10$, $H_0 = 95.5$, and $H_1 = .95$. Express the solution of the firm's problem in the form

$$y_{t+1} = h_0 + h_1 y_t + h_2 Y_t, \quad (5)$$

giving values for the h_j 's.

- d. If there were n identical competitive firms all behaving according to equation (5), what would equation (5) imply for the *actual* law of motion (4) for the market supply Y ?
- e. Formulate the Euler equation for the firm's problem.

Solution:

(a) Substitute (3) into (2): $R_t = A_0 y_t - A_1 Y_t y_t - .5d(y_{t+1} - y_t)^2$. The firm controls y_{t+1} , knows (y_t, Y_t) , and forecasts Y with (4). Its Bellman equation is (7.2.5):

$$v(y, Y) = \max_{y'} \left\{ A_0 y - A_1 Y y - .5d(y' - y)^2 + \beta v(y', Y') \right\} \quad \text{s.t. } Y' = H_0 + H_1 Y.$$

(b) The firm needs y (its own output), Y (to know today's price and forecast tomorrow's) and a constant (for the intercepts A_0, H_0). So the **state** is $x_t = [y_t, Y_t, 1]'$. Take the control to be the change in output, $u_t = y_{t+1} - y_t$. Then

$$x_{t+1} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & H_1 & H_0 \\ 0 & 0 & 1 \end{bmatrix}}_A x_t + \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_B u_t.$$

The return must be written as $R_t = -(x_t' R x_t + u_t' Q u_t)$. With

$$R = \begin{bmatrix} 0 & A_1/2 & -A_0/2 \\ A_1/2 & 0 & 0 \\ -A_0/2 & 0 & 0 \end{bmatrix}, \quad Q = \frac{d}{2},$$

we get $x' R x = 2 \cdot \frac{A_1}{2} y Y + 2 \cdot \left(-\frac{A_0}{2}\right) y = A_1 y Y - A_0 y$ and $u' Q u = .5 d u^2$, so $-(x' R x + u' Q u) = A_0 y - A_1 Y y - .5 d u^2 = R_t$. ✓ The objects are (β, A, B, R, Q) ; the firm maximises $-\sum_t \beta^t \{x_t' R x_t + u_t' Q u_t\}$ given $x_0 = [y_0, Y_0, 1]'$, with solution $v(x) = -x' P x$, $u_t = -F x_t$. *Remark.* R is not positive semidefinite: revenue is *linear* in y . The problem is still well posed. With $B = e_1$, $B' P B = P_{yy}$ and the (y, y) entry of the Riccati map is $R_{yy} + \beta P_{yy} - \beta^2 (P_{yy})^2 / (Q + \beta P_{yy})$. Starting from $P = 0$ it stays at $P_{yy} = 0$ at every iteration, so v is linear in y . The discounted system is stable because the roots of $A - B F$ are at most $1 < 1/\sqrt{\beta}$.

(c) From $u = -F x$, $y_{t+1} = y_t + u_t = (1 - F_1) y_t - F_2 Y_t - F_3$, so $h_0 = -F_3$, $h_1 = 1 - F_1$, $h_2 = -F_2$. Iterating the Riccati equation from $P = 0$ (olrp, 489 steps) gives

$$\boxed{y_{t+1} = 96.9487 + 1.0000 y_t - 0.046282 Y_t,}$$

$$P = \begin{bmatrix} 0 & 0.25641 & -534.744 \\ 0.25641 & -0.07509 & 163.721 \\ -534.744 & 163.721 & -358775.96 \end{bmatrix}.$$

Part (e) derives the same numbers in closed form: $h_1 = 1$, $h_2 = -\beta A_1 H_1 / [d(1 - \beta H_1)] = -.045125 / .975 = -0.0462821$, $h_0 = \beta(A_0 - A_1 H_0 + d h_2 H_0) / [d(1 - \beta)] = .95(100 - 4.775 - 44.1994) / .5 = 96.9487$.

Reading of $h_1 = 1$. The firm is a price taker and revenue is linear in y , so its own level of output does not affect how fast it wants to change it. The change $y_{t+1} - y_t = 96.95 - 0.0463 Y_t$ depends only on the forecast of prices, i.e. on Y_t .

(d) With n identical firms, $Y_t = n y_t$ and each firm uses (5). Multiply (5) by n :

$$Y_{t+1} = n y_{t+1} = n h_0 + h_1 (n y_t) + n h_2 Y_t \implies \boxed{Y_{t+1} = n h_0 + (h_1 + n h_2) Y_t,}$$

which is (7.2.10) for this model. So the actual law has $H_0^A = n h_0$ and $H_1^A = h_1 + n h_2$. For $n = 1$: $Y_{t+1} = 96.9487 + 0.953718 Y_t \neq 95.5 + .95 Y_t$. The beliefs of part (c) are

not rational: firms that hold them act in a way that falsifies them. (For $n = 10$: $Y_{t+1} = 969.487 + 0.537179 Y_t$.)

(e) First-order condition for y' in (a): $-d(y' - y) + \beta v_y(y', Y') = 0$ (7.2.6). The envelope (Benveniste–Scheinkman) condition, differentiating (a) with respect to y at the optimum: $v_y(y, Y) = A_0 - A_1 Y + d(y' - y)$. Shift it forward one period and substitute:

$$-d(y_{t+1} - y_t) + \beta[A_0 - A_1 Y_{t+1} + d(y_{t+2} - y_{t+1})] = 0,$$

which is (7.2.7). Here $Y_{t+1} = H_0 + H_1 Y_t$, and the terminal condition is $\lim_t \beta^t y_t v_y(y_t, Y_t) = 0$ (7.2.8).

Solving it (the closed form used in (c)). Write $u_t = y_{t+1} - y_t$ and $p_t = A_0 - A_1 Y_t$. The Euler equation reads $du_t = \beta p_{t+1} + \beta du_{t+1}$. Its homogeneous part $d\beta\lambda^2 - d(1+\beta)\lambda + d = 0$ in y has roots $\lambda = 1$ and $\lambda = 1/\beta$. The transversality condition rules out the root $1/\beta$, so we solve forward:

$$u_t = \frac{1}{d} \sum_{j \geq 0} \beta^{j+1} p_{t+1+j}.$$

The firm changes output by the discounted value of all future prices divided by d . Under (4), $Y_{t+1+j} = H_1^{j+1} Y_t + H_0 \sum_{i=0}^j H_1^i$, so the coefficient of Y_t in u_t is $-\frac{A_1}{d} \sum_{j \geq 0} (\beta H_1)^{j+1} = -\frac{\beta A_1 H_1}{d(1-\beta H_1)} = h_2$.

By undetermined coefficients. Guess $y_{t+1} = h_0 + h_1 y_t + h_2 Y_t$. Then $y_{t+2} - y_{t+1} = h_0 + (h_1 - 1)y_{t+1} + h_2 Y_{t+1}$. Substitute into (7.2.7) and match coefficients:

$$y_t: \quad -d(h_1 - 1) + \beta d(h_1 - 1)h_1 = 0 \Rightarrow (h_1 - 1)(\beta h_1 - 1) = 0 \Rightarrow h_1 = 1,$$

$$Y_t: \quad -dh_2 - \beta A_1 H_1 + \beta dh_2 H_1 = 0 \Rightarrow h_2 = -\frac{\beta A_1 H_1}{d(1 - \beta H_1)},$$

$$1: \quad -dh_0 + \beta A_0 - \beta A_1 H_0 + \beta dh_0 + \beta dh_2 H_0 = 0 \\ \Rightarrow h_0 = \frac{\beta(A_0 - A_1 H_0 + dh_2 H_0)}{d(1 - \beta)}.$$

(The root $h_1 = 1/\beta$ violates (7.2.8). With $h_1 = 1$ the y_{t+1} terms cancel: $y_{t+2} - y_{t+1} = h_0 + h_2 Y_{t+1}$.)

Numerical exercise — see the companion code

`olrp`, `scipy`'s `DARE` and the closed form agree to 10^{-9} . The quadratic form matches R_t at random points. The Euler residual of (7.2.7) along a 60-period path generated by (5) and (4) is 6×10^{-11} . The computed P has $P_{yy} = 0$ exactly.

Exercise 7.2 Rational expectations

Question

p. 245. Now assume that the firm in problem 1 is “representative.” We implement this idea by setting $n = 1$. In equilibrium, we will require that $y_t = Y_t$, but we don’t want to impose this condition at the stage that the firm is optimizing (because we want to retain competitive behavior). Define a rational expectations equilibrium to be a pair of numbers H_0, H_1 such that if the representative firm solves the problem ascribed to it in problem 1, then the firm’s optimal behavior given by equation (5) implies that $y_t = Y_t \forall t \geq 0$.

a. Use the program that you wrote for exercise 7.1 to determine which if any of the following pairs (H_0, H_1) is a rational expectations equilibrium: (i) (94.0888, .9211); (ii) (93.22, .9433), and (iii) (95.08187459215024, .95245906270392)?

b. Describe an iterative algorithm that uses the program that you wrote for exercise 7.1 to compute a rational expectations equilibrium. (You are not being asked actually to use the algorithm you are suggesting.)

Solution:

(a) By 7.1(d) with $n = 1$, beliefs $H = (H_0, H_1)$ lead to the actual law $\mathcal{M}(H) = (h_0(H), h_1(H) + h_2(H))$. We need $y_t = Y_t$ for all t when $y_0 = Y_0$. By induction this holds iff $h_0 + h_1 Y + h_2 Y = H_0 + H_1 Y$ for all Y , i.e. iff $\mathcal{M}(H) = H$. Running the program of 7.1 at each candidate:

	perceived (H_0, H_1)	actual $(h_0, h_1 + h_2)$	REE?
(i)	(94.0888, .9211)	(118.46676, 0.9649856)	no
(ii)	(93.22, .9433)	(104.73644, 0.9568606)	no
(iii)	(95.081875, .952459)	(95.081875, 0.952459)	yes

Only **(iii)** is a rational expectations equilibrium; its gap is 1.3×10^{-10} , which is the tolerance of the Riccati iteration.

Closed form (independent check). Use the formulas of 7.1(e). The slope condition $H_1 = 1 + h_2(H_1)$ reads

$$\begin{aligned}
 H_1 - 1 &= -\frac{\beta A_1 H_1}{d(1 - \beta H_1)} \iff d(H_1 - 1)(1 - \beta H_1) + \beta A_1 H_1 = 0 \\
 &\iff d\beta H_1^2 - [d(1 + \beta) + \beta A_1]H_1 + d = 0.
 \end{aligned}$$

With our numbers this is $9.5H_1^2 - 19.5475H_1 + 10 = 0$. The roots are 0.95245906 and 1.10517; their product is $1/\beta$. The stable root is $H_1 = 0.95245906270392$. The intercept condition $H_0 = h_0(H_0, H_1)$ with $dh_2 = d(H_1 - 1)$ gives

$$\begin{aligned}
 H_0 d(1 - \beta) &= \beta [A_0 - A_1 H_0 - d(1 - H_1)H_0] \implies H_0 = \frac{\beta A_0}{d(1 - \beta) + \beta A_1 + \beta d(1 - H_1)}, \\
 H_0 &= \frac{95}{0.5 + 0.0475 + 0.4516389} = 95.0818746.
 \end{aligned}$$

The implied steady state is $H_0/(1 - H_1) = 2000 = A_0/A_1$. There the price is zero: with no production cost other than adjustment, a steady state needs $u = 0$, and by the forward solution of 7.1(e) that requires $\sum \beta^j p = 0$.

(b) *Algorithm (iterate on the map from perceived to actual laws).*

1. Guess $H^{(0)} = (H_0^{(0)}, H_1^{(0)})$, e.g. (95.5, .95); choose a relaxation weight $\gamma \in (0, 1]$ and a tolerance ε .
2. Given $H^{(j)}$, build (A, B, R, Q) of 7.1(b) with $(H_0, H_1) = H^{(j)}$. Solve it with the program of 7.1 and read off (h_0, h_1, h_2) .
3. Form the actual law $\mathcal{M}(H^{(j)}) = (nh_0, h_1 + nh_2)$ with $n = 1$.
4. Update $H^{(j+1)} = (1 - \gamma)H^{(j)} + \gamma \mathcal{M}(H^{(j)})$.
5. Stop when $\|H^{(j+1)} - H^{(j)}\| < \varepsilon$; then $H^{(j+1)}$ is (approximately) an REE. Otherwise return to step 2.

Why the relaxation γ . $\mathcal{M}$ is not a contraction in general (footnote 5, p. 228). Near the fixed point its Jacobian is triangular, because $h_1 + h_2$ does not depend on H_0 . The diagonal entries are

$$\frac{\partial h_0}{\partial H_0} = \frac{\beta(-A_1 + dh_2)}{d(1 - \beta)} = \frac{\beta[-A_1 - d(1 - H_1)]}{d(1 - \beta)} = -0.99828,$$

$$\frac{\partial(1 + h_2)}{\partial H_1} = -\frac{\beta A_1}{d(1 - \beta H_1)^2} = -0.52451.$$

With $\gamma = 1$ the error in H_0 shrinks by the factor 0.998 per step and flips sign each time, so convergence is extremely slow. With $\gamma = .5$ the eigenvalues become $1 - \gamma + \gamma\lambda$, i.e. 0.0009 and 0.238, and convergence is fast. Part (b) does not ask for this, but running the algorithm confirms it: after 150 steps from (95.5, .95) the error is 3.4 with $\gamma = 1$ and 9×10^{-11} with $\gamma = .5$.

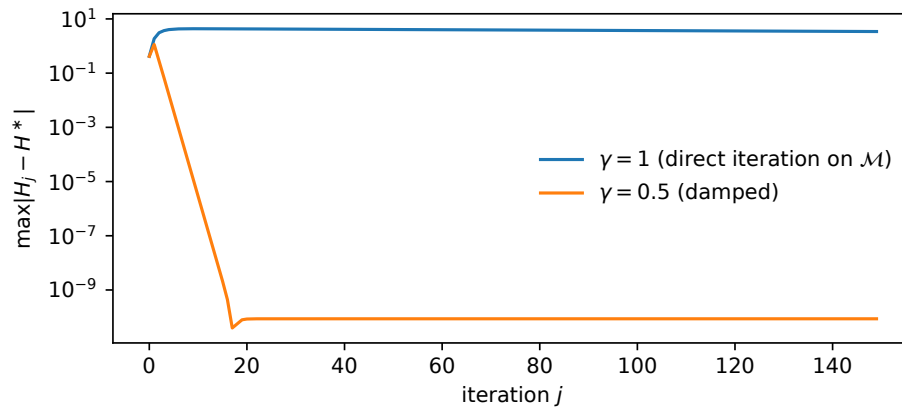


Figure 1: Exercise 7.2(b). Distance to the REE by iteration on the map from perceived to actual laws. Each step calls the 7.1 program once.

Numerical exercise — see the companion code

The fixed point from the quadratic equals pair (iii) to 10^{-10} . Central finite differences of $\mathcal{M}$ (each evaluation one `olrp` call) match the two closed-form Jacobian entries to 10^{-5} .

Exercise 7.3 Maximizing welfare*Question*

p. 246. A planner seeks to maximize the welfare criterion

$$\sum_{t=0}^{\infty} \beta^t S_t, \quad (1)$$

where S_t is “consumer surplus plus producer surplus” defined to be

$$S_t = S(Y_t, Y_{t+1}) = \int_0^{Y_t} (A_0 - A_1 x) dx - .5d(Y_{t+1} - Y_t)^2.$$

- Formulate the planner’s Bellman equation.
- Formulate the planner’s problem as an optimal linear regulator, and, for the same parameter values in exercise 7.1, solve it using the Matlab program `olrp.m`. Represent the solution in the form $Y_{t+1} = s_0 + s_1 Y_t$.
- Compare your answer in part b with your answer to part a of exercise 7.2.

Solution:

(a) $\int_0^Y (A_0 - A_1 x) dx = A_0 Y - \frac{A_1}{2} Y^2$. The only state is Y ; the planner chooses Y' . The Bellman equation is (7.2.13):

$$V(Y) = \max_{Y'} \left\{ A_0 Y - \frac{A_1}{2} Y^2 - .5d(Y' - Y)^2 + \beta V(Y') \right\}.$$

(b) State $x_t = [Y_t, 1]'$, control $u_t = Y_{t+1} - Y_t$, $A = I_2$, $B = [1, 0]'$,

$$R = \begin{bmatrix} A_1/2 & -A_0/2 \\ -A_0/2 & 0 \end{bmatrix}, \quad Q = \frac{d}{2}.$$

Check: $x'Rx = \frac{A_1}{2}Y^2 - A_0Y$, so $-(x'Rx + u'Qu) = A_0Y - \frac{A_1}{2}Y^2 - .5du^2 = S_t$. ✓ Then $Y_{t+1} = (1 - F_1)Y_t - F_2$. `olrp` (496 steps) gives

$$Y_{t+1} = 95.08187459215 + 0.95245906270392 Y_t, \quad P = \begin{bmatrix} 0.2627 & -525.409 \\ -525.409 & -949181.25 \end{bmatrix}.$$

(c) (s_0, s_1) is *identical* to pair (iii) of 7.2(a), the rational expectations equilibrium. The reason is the argument of section 7.2.1. The first-order condition of (a) is $-d(Y' - Y) +$

$\beta V'(Y') = 0$, and the envelope condition is $V'(Y) = A_0 - A_1 Y + d(Y' - Y)$. Combining them:

$$\beta A_0 + dY_t - [\beta A_1 + d(1 + \beta)]Y_{t+1} + d\beta Y_{t+2} = 0 \quad (7.2.16).$$

Setting $y_t = Y_t$ in the firm's Euler equation (7.2.7) gives the same equation: $-dY_{t+1} + dY_t + \beta A_0 - \beta A_1 Y_{t+1} + \beta dY_{t+2} - \beta dY_{t+1} = 0$. Its characteristic polynomial $d\beta z^2 - [\beta A_1 + d(1 + \beta)]z + d$ is the quadratic of 7.2(a). The stable root is $s_1 = H_1$, and the steady state $Y = A_0/A_1$ gives $s_0 = (1 - s_1)A_0/A_1 = H_0$. The competitive equilibrium is efficient; this is the first welfare theorem, and the equilibrium price $p_t = A_0 - A_1 Y_t$ is the planner's shadow price of output.

Numerical exercise — see the companion code

S_t by numerical quadrature equals the quadratic form at random points. `olrp` and `DARE` agree to 10^{-8} , and (s_0, s_1) equals the 7.2 closed form to 10^{-8} .

Market Power: Monopoly and Duopoly

Exercise 7.4 Monopoly

Question

p. 246. A monopolist faces the industry demand curve (3) and chooses Y_t to maximize $\sum_{t=0}^{\infty} \beta^t R_t$ where $R_t = p_t Y_t - .5d(Y_{t+1} - Y_t)^2$ and where Y_0 is given.

- Formulate the firm's Bellman equation.
- For the parameter values listed in exercise 7.1, formulate and solve the firm's problem using `olrp.m`.
- Compare your answer in part b with the answer you obtained to part b of exercise 7.3.

Solution:

(a) The monopolist knows that its own output sets the price: $p_t Y_t = A_0 Y_t - A_1 Y_t^2$. There is no aggregate variable to forecast, hence no perceived law:

$$V(Y) = \max_{Y'} \left\{ A_0 Y - A_1 Y^2 - .5d(Y' - Y)^2 + \beta V(Y') \right\}.$$

(b) This is the planner's regulator of 7.3(b) with $A_1/2$ replaced by A_1 : $x = [Y, 1]'$, $u = Y' - Y$, $A = I_2$, $B = [1, 0]'$, $Q = d/2$,

$$R = \begin{bmatrix} A_1 & -A_0/2 \\ -A_0/2 & 0 \end{bmatrix} \quad (x' R x = A_1 Y^2 - A_0 Y).$$

olrp gives

$$Y_{t+1} = 73.472944 + 0.926527 Y_t, \quad P = \begin{bmatrix} 0.41736 & -417.365 \\ -417.365 & -582635.28 \end{bmatrix}.$$

Check by the Euler equation. The FOC $-d(Y' - Y) + \beta V'(Y') = 0$ and the envelope condition $V'(Y) = A_0 - 2A_1Y + d(Y' - Y)$ give $\beta A_0 + dY_t - [2\beta A_1 + d(1 + \beta)]Y_{t+1} + d\beta Y_{t+2} = 0$. The characteristic equation $9.5z^2 - 19.595z + 10 = 0$ has roots 0.926527 and 1.136106; the stable root is s_1 . The steady state solves $A_0 - 2A_1Y = 0$, so $Y = A_0/(2A_1) = 1000$ and $s_0 = (1 - 0.926527) \times 1000 = 73.4729$. ✓

(c) The monopolist solves the planner's problem with the demand curve replaced by the *marginal revenue* curve $A_0 - 2A_1Y$, i.e. with slope $2A_1$ instead of A_1 .

	$Y_{t+1} = s_0 + s_1 Y_t$	steady-state Y	steady-state p
planner (= competition, 7.3)	$95.0819 + 0.952459 Y_t$	2000	0
monopoly (7.4)	$73.4729 + 0.926527 Y_t$	1000	50

The monopolist restricts output to half the efficient level and keeps the price at 50. That is a permanent deadweight loss (footnote 9, p. 238): the market outcome no longer maximises $\sum \beta^t S_t$. Its adjustment root is smaller ($0.9265 < 0.9525$), so it moves faster toward a target that is closer. The steeper marginal-revenue curve raises the cost of being away from the target relative to the adjustment cost d .

Numerical exercise — see the companion code

olrp, DARE and the Euler-equation closed form agree to 10^{-8} .

Exercise 7.5 Duopoly

Question

pp. 246–247. An industry consists of two firms that jointly face the industry-wide inverse demand curve $p_t = A_0 - A_1 Y_t$, where now $Y_t = y_{1t} + y_{2t}$. Firm $i = 1, 2$ maximizes

$$\sum_{t=0}^{\infty} \beta^t R_{it} \quad (1)$$

where $R_{it} = p_t y_{it} - .5d(y_{i,t+1} - y_{it})^2$.

- Define a Markov perfect equilibrium for this industry.
- Formulate the Bellman equation for each firm.
- Use the Matlab program `nash.m` to compute an equilibrium, assuming the parameter values listed in exercise 7.1.

Solution: Substituting demand, $R_{it} = A_0 y_{it} - A_1 y_{it}^2 - A_1 y_{it} y_{-i,t} - .5d(y_{i,t+1} - y_{it})^2$ (7.6.3).

(a) Firm i 's strategy is a decision rule $y_{i,t+1} = f_i(y_{it}, y_{-i,t})$ that depends only on the

payoff-relevant state (y_{1t}, y_{2t}) . A **Markov perfect equilibrium** is a pair of value functions v_1, v_2 and a pair of policy functions f_1, f_2 such that, for $i = 1, 2$: (i) given f_{-i} , v_i satisfies firm i 's Bellman equation in (b) for *every* state (y_i, y_{-i}) ; and (ii) $f_i(y_i, y_{-i})$ attains the maximum on its right side. “Markov” means that strategies depend on the current state only, not on history. “Perfect” means that optimality holds at all states, including those never reached on the equilibrium path, so the equilibrium is built by backward induction (p. 239). In the linear-quadratic version sought in (c), f_i is affine: $u_{it} = -F_i x_t$ with $x_t = [1, y_{1t}, y_{2t}]'$ (section 7.7).

(b) For $i = 1, 2$ (7.6.4)–(7.6.5):

$$\boxed{v_i(y_i, y_{-i}) = \max_{y'_i} \left\{ A_0 y_i - A_1 y_i^2 - A_1 y_i y_{-i} - .5d(y'_i - y_i)^2 + \beta v_i(y'_i, y'_{-i}) \right\} \\ \text{s.t. } y'_{-i} = f_{-i}(y_{-i}, y_i).$$

Unlike the competitive firm of 7.1, firm i knows that its rival's next output responds to y_i through f_{-i} , and that its own output moves the price.

(c) Map the game into the form solved by **nnash** (p. 243). State $x_t = [1, y_{1t}, y_{2t}]'$, controls $u_{it} = y_{i,t+1} - y_{it}$, law of motion $x_{t+1} = Ax_t + B_1 u_{1t} + B_2 u_{2t}$ with $A = I_3$, $B_1 = e_2$, $B_2 = e_3$. Player i maximises $-\sum \beta^t \{x'_t R_i x_t + u'_{it} Q_i u_{it}\}$ with $Q_i = d/2$,

$$R_1 = \begin{bmatrix} 0 & -A_0/2 & 0 \\ -A_0/2 & A_1 & A_1/2 \\ 0 & A_1/2 & 0 \end{bmatrix}, \quad R_2 = \begin{bmatrix} 0 & 0 & -A_0/2 \\ 0 & 0 & A_1/2 \\ -A_0/2 & A_1/2 & A_1 \end{bmatrix}.$$

Check: $x' R_1 x = -A_0 y_1 + A_1 y_1^2 + A_1 y_1 y_2$, so $-(x' R_1 x + u'_1 Q_1 u_1) = R_{1t}$. ✓ (The book's s_i, w_i, m_i are zero here.)

Algorithm (7.7.4)–(7.7.7) with discounting. Start from $P_1 = P_2 = 0$. Given (P_1, P_2) , the pair (F_1, F_2) solves the linear system

$$\begin{aligned} (Q_1 + \beta B'_1 P_1 B_1) F_1 + \beta B'_1 P_1 B_2 F_2 &= \beta B'_1 P_1 A, \\ \beta B'_2 P_2 B_1 F_1 + (Q_2 + \beta B'_2 P_2 B_2) F_2 &= \beta B'_2 P_2 A. \end{aligned}$$

This is (7.7.4) and (7.7.6) written jointly: firm i 's best response to F_{-i} in the one-more-period game. Then update $P_i \leftarrow R_i + F'_i Q_i F_i + \beta(A - B_1 F_1 - B_2 F_2)' P_i (A - B_1 F_1 - B_2 F_2)$, which is (7.7.5) and (7.7.7) evaluated at the chosen F_i . Iterate until (P_1, P_2) converge. This took 487 steps and gave

$$F_1 = [-65.27422, 0.06708, 0.02423], \quad F_2 = [-65.27422, 0.02423, 0.06708],$$

so that

$$\boxed{y_{i,t+1} = 65.27422 + 0.93292 y_{it} - 0.02423 y_{-i,t}, \quad i = 1, 2.}$$

The closed-loop matrix $A - B_1 F_1 - B_2 F_2$ has roots 1 (the constant), 0.95716 and 0.90869. The symmetric steady state is $y_i = 65.27422 / (1 - 0.93292 + 0.02423) = 714.90$. Industry

output is 1429.80 and the price is 28.51. Starting from $y_{10} = y_{20}$, industry output follows $Y_{t+1} = 130.548 + 0.908695 Y_t$.

Comparison. Steady-state industry output is 1000 under monopoly, 1333.3 under static Cournot ($2A_0/3A_1$), 1429.8 in the dynamic duopoly and 2000 under competition. The dynamic duopolists produce *more* than static Cournot firms. The coefficient -0.02423 says that a firm's current output lowers its rival's next-period output. Output is like a capital stock that deters the rival, so each firm over-expands to get this strategic effect. The effect is absent from the static game.

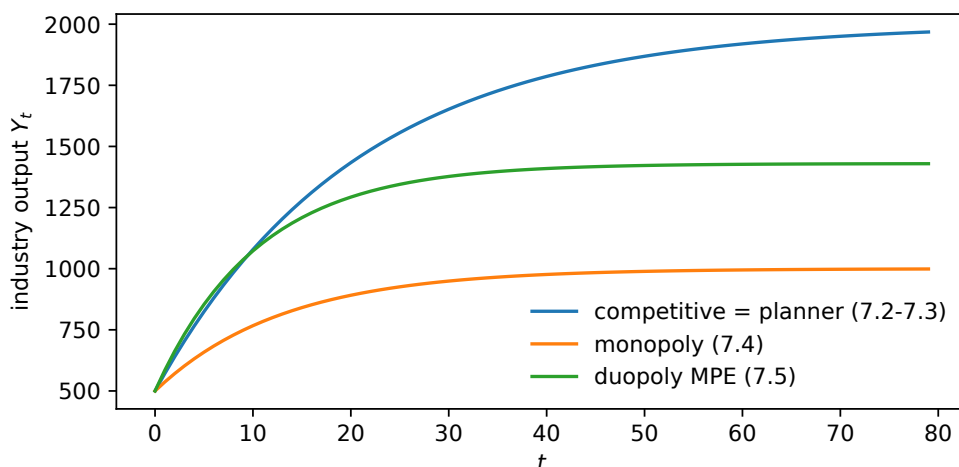


Figure 2: Exercises 7.2–7.5. Industry output from $Y_0 = 500$ (duopoly: $y_{i0} = 250$).

Numerical exercise — see the companion code

Independent route: best-response iteration. Solve firm 1's infinite-horizon regulator given F_2 (state law $A - B_2 F_2$, via `olrp`), then firm 2's given the new F_1 , and repeat. It converges in 12 rounds to the **nnash** pair to 10^{-7} . `scipy`'s DARE on firm 1's problem given F_2 returns F_1 and P_1 . The rules are symmetric. The statement names **nash.m**; the program supplied with the book is **nnash.m** (p. 243).

Markov Perfection Beyond Linear Games

Exercise 7.6 Self-control

Question

p. 247. This is a model of a human who has time inconsistent preferences, of a type proposed by Phelps and Pollak (1968) and used by Laibson (1994). The human lives from $t = 0, \dots, T$. Think of the human as actually consisting of $T + 1$ personalities, one for each period. Each personality is a distinct agent (i.e., a distinct utility function and constraint set). Personality T has preferences ordered by $u(c_T)$ and personality $t < T$ has preferences that are ordered by

$$u(c_t) + \delta \sum_{j=1}^{T-t} \beta^j u(c_{t+j}),$$

where $u(\cdot)$ is a twice continuously differentiable, increasing, and strictly concave function of consumption of a single good; $\beta \in (0, 1)$, and $\delta \in (0, 1]$. When $\delta < 1$, preferences of the sequence of personalities are time inconsistent (that is, not recursive). At each t , let there be a savings technology described by

$$k_{t+1} + c_t \leq f(k_t),$$

where f is a production function with $f' > 0$, $f'' \leq 0$.

- a. Define a Markov perfect equilibrium for the $T + 1$ personalities.
- b. Argue that the Markov perfect equilibrium can be computed by iterating on the following functional equations:

$$\begin{aligned} V_{j+1}(k) &= \max_c \{u(c) + \beta \delta W_j(k')\} \\ W_{j+1}(k) &= u[c_{j+1}(k)] + \beta W_j[f(k) - c_{j+1}(k)] \end{aligned}$$

where $c_{j+1}(k)$ is the maximizer of the right side of the first equation above for $j + 1$, starting from $W_0(k) = u[f(k)]$. Here $W_j(k)$ is the value of $u(c_{T-j}) + \beta u(c_{T-j+1}) + \dots + \beta^{T-j} u(c_T)$, taking the decision rules $c_h(k)$ as given for $h = 0, 1, \dots, j$.

- c. State the optimization problem of the time 0 person who is given the power to dictate the choices of all subsequent persons. Write the Bellman equations for this problem. The time 0 person is said to have a commitment technology for “self-control” in this problem.

Solution: Index rules by *periods to go*. $c_h(k)$ is the rule of personality $T - h$, so c_0 belongs to personality T . This is the indexing the functional equations use.

Misprint. The last term of W_j should be $\beta^j u(c_T)$, not $\beta^{T-j} u(c_T)$. W_j starts at date $T - j$, so date T is j periods later. The recursion $W_{j+1} = u + \beta W_j$ with $W_0 = u[f(k)]$ produces exactly β^j .

(a) The payoff-relevant state at t is k_t . A **Markov perfect equilibrium** is a collection of consumption rules $\{c_h(k)\}_{h=0}^T$, one per personality, with $0 \leq c_h(k) \leq f(k)$, such that

for every h and *every* k , $c_h(k)$ maximises personality $T - h$'s objective

$$u(c) + \delta \sum_{i=1}^h \beta^i u(c_{T-h+i}),$$

where the future is generated by $k_{T-h+1} = f(k) - c$ and, for $i \geq 1$, by the successors' rules: $c_{T-h+i} = c_{h-i}(k_{T-h+i})$ and $k_{T-h+i+1} = f(k_{T-h+i}) - c_{T-h+i}$. Each personality takes its successors' rules as given (Nash), strategies depend only on k (Markov), and optimality holds at all k , on or off the path (perfect). Personality T has no future, so $c_0(k) = f(k)$.

(b) Backward induction.

Start. Personality T consumes everything: $c_0(k) = f(k)$ and $V_0(k) = W_0(k) = u[f(k)]$.

Step. Suppose the rules $c_0, \dots, c_j$ of personalities $T, \dots, T - j$ are known, and let $W_j(k)$ be the β -discounted utility from date $T - j$ onward that they generate from $k_{T-j} = k$: $W_j(k) = \sum_{i=0}^j \beta^i u(c_{T-j+i})$. Personality $t = T - j - 1$ (with $j + 1$ periods to go) values

$$u(c_t) + \delta \sum_{i=1}^{j+1} \beta^i u(c_{t+i}) = u(c_t) + \delta \beta \sum_{i=0}^j \beta^i u(c_{T-j+i}) = u(c_t) + \beta \delta W_j(k_{t+1}).$$

The first equality factors out $\delta \beta$ and re-indexes $i \rightarrow i + 1$. The second uses the fact that, given the successors' Markov rules, the continuation depends only on $k_{t+1} = f(k) - c_t$. So personality t solves

$$V_{j+1}(k) = \max_c \{u(c) + \beta \delta W_j(f(k) - c)\},$$

the first functional equation, and its rule is the maximiser $c_{j+1}(k)$. Now extend W one period:

$$\begin{aligned} W_{j+1}(k) &= \sum_{i=0}^{j+1} \beta^i u(c_{T-j-1+i}) = u[c_{j+1}(k)] + \beta \sum_{i=0}^j \beta^i u(c_{T-j+i}) \\ &= u[c_{j+1}(k)] + \beta W_j[f(k) - c_{j+1}(k)], \end{aligned}$$

the second functional equation. Iterating $j = 0, \dots, T - 1$ produces every rule. Each is optimal for its personality at every k given its successors, which is the definition in (a).

Why two functions. V_{j+1} is the current personality's own value; it weights tomorrow by $\beta \delta$. W_{j+1} is how that personality's predecessors value the same future; it weights tomorrow by β , because δ is common to all future dates from their viewpoint. Hence $V_{j+1}(k) = W_{j+1}(k) - \beta(1 - \delta)W_j(k')$ at $k' = f(k) - c_{j+1}(k)$. The W update evaluates c_{j+1} ; it does *not* maximise. So the envelope theorem does not apply to W and W_j need not be concave when $\delta < 1$. The argmax may then be set-valued and a selection is needed; the recursion still defines an MPE. With $\delta = 1$, $V = W$ and the scheme is ordinary backward dynamic programming.

(c) With commitment, the time-0 person chooses the whole plan:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^T} u(c_0) + \delta \sum_{t=1}^T \beta^t u(c_t) \quad \text{s.t.} \quad k_{t+1} + c_t \leq f(k_t), \quad k_{t+1} \geq 0, \quad k_0 \text{ given.}$$

Write the objective as $u(c_0) + \delta\beta\left[\sum_{t=1}^T \beta^{t-1}u(c_t)\right]$. From date 1 on, the bracket is an ordinary geometric criterion, so the plan from date 1 is time consistent *for person 0*. The Bellman equations are, with J_h the value with h periods to go,

$$\begin{aligned} J_0(k) &= u[f(k)], \\ J_{h+1}(k) &= \max_c \{u(c) + \beta J_h(f(k) - c)\}, \quad h = 0, \dots, T-2, \\ V^c(k_0) &= \max_{c_0} \{u(c_0) + \beta\delta J_{T-1}(f(k_0) - c_0)\}. \end{aligned}$$

Only the first step carries $\beta\delta$; every later step uses β . In the MPE, every personality applies $\beta\delta$ to its own tomorrow. That is the self-control problem: person 0 would like persons $1, \dots, T-1$ to use β .

Worked example (parameters chosen: $u = \ln c$, $f(k) = \bar{A}k^\alpha$, $\bar{A} = 1$, $\alpha = .36$, $\beta = .96$, $\delta = .6$, $T = 10$). Guess $W_j(k) = a_j \ln k + b_j$. $W_0 = \ln \bar{A} + \alpha \ln k$, so $a_0 = \alpha$. Personality with $j+1$ periods to go: $\max_c \{\ln c + \beta\delta a_j \ln(f - c)\}$. The FOC $1/c = \beta\delta a_j/(f - c)$ gives

$$c_{j+1}(k) = \frac{f(k)}{1 + \beta\delta a_j}, \quad k' = \frac{\beta\delta a_j}{1 + \beta\delta a_j} f(k).$$

Then $W_{j+1}(k) = \ln c + \beta(a_j \ln k' + b_j) = (1 + \beta a_j) \ln f(k) + \text{const}$, so

$$a_{j+1} = \alpha(1 + \beta a_j) \rightarrow \frac{\alpha}{1 - \alpha\beta}.$$

The coefficient recursion does not involve δ . With log utility, consumption and savings are proportional to $f(k)$, so the saving *rate* affects only the constants b_j . Commitment gives $J_h = \tilde{a}_h \ln k + \dots$ with the same recursion, so $\tilde{a}_h = a_h$. Hence:

saving rate $k_{t+1}/f(k_t)$ at date $t < T$	MPE	commitment
$t = 0$	$\frac{\beta\delta a_{T-1}}{1 + \beta\delta a_{T-1}}$	same
$1 \leq t \leq T-1$	$\frac{\beta\delta a_{T-t-1}}{1 + \beta\delta a_{T-t-1}}$	$\frac{\beta a_{T-t-1}}{1 + \beta a_{T-t-1}}$

Numbers: $a = (0.36, 0.4844, 0.5274, \dots, 0.5501)$. The MPE saving rates are $0.2406, 0.2406, 0.2406, 0.2405, 0.2403, 0.2397, 0.2380, 0.2330, 0.2182, 0.1718$ for $t = 0, \dots, 9$. The commitment rates are $0.2406, 0.3456, 0.3456, 0.3455, 0.3452, 0.3445, 0.3424, 0.3361, 0.3174, 0.2568$. From $k_0 = .2$, person 0's welfare is -5.8944 in the MPE and -5.7140 under commitment.

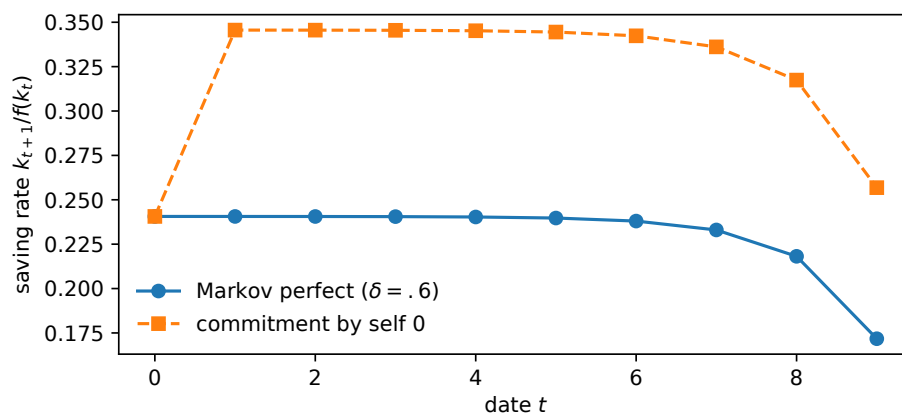


Figure 3: Exercise 7.6. Without commitment each future self saves less than self 0 wants.

Numerical exercise — see the companion code

The functional equations of (b), and the commitment Bellman equations of (c), are iterated on a 120-point grid in $\ln k$. W_j is interpolated by cubic spline and the maximisation is numerical (bounded scalar search over the saving rate). All saving rates match the closed form to 10^{-6} at every grid point. The notebook also checks that the welfare of self 0 is higher under commitment.

Exercise 7.7 Equilibrium search

Question

p. 248. An economy consists of a continuum of *ex ante* identical workers each of whom is either employed or unemployed. A worker wants to maximize the expected value of $\sum_{t=0}^{\infty} \beta^t y_t$ where $\beta \in (0, 1)$ and

$$y_t = \begin{cases} w & \text{if employed} \\ c(U) & \text{if unemployed.} \end{cases}$$

Each period, an unemployed worker draws one and only one offer to work (until fired) at a wage w drawn from a c.d.f. $F(W) = \text{Prob}(w \leq W)$ where $F(0) = 0$, $F(B) = 1$ for $B > 0$. Successive draws from F are i.i.d. If a worker accepts a job, he receives w this period and enters the beginning of next period as ‘employed’. At the beginning of each period, each such previously employed worker is exposed to a probability of $\lambda \in (0, 1)$ of being fired; with probability $1 - \lambda$ he is not fired and again receives the previously drawn w as a wage. If fired, the worker becomes newly unemployed and has the same opportunity as all other unemployed workers, i.e., he draws an offer w from c.d.f. F . If an unemployed worker rejects that offer, he receives unemployment compensation $c(U) = c\left[\frac{1}{1+\exp(-6U)} - .5\right]$ and enters next period unemployed. Here U is the aggregate unemployment rate at the beginning of the period. The unemployment rate tomorrow U^* is related to the unemployment rate U today by the law of motion

$$U^* = \lambda(1 - U) + (1 - \phi(U))U,$$

where $\phi(U)$ is the fraction of unemployed workers who accept a wage offer this period.

- a. Write a Bellman equation for an unemployed worker.
- b. Describe the form of an unemployed worker’s optimal decision rule.
- c. Describe how $\phi(U)$ is implied by a typical worker’s optimal decision rule.
- d. Define a recursive competitive equilibrium for this environment.

Solution: Benefits depend on the aggregate unemployment rate, so a worker must forecast U . This is the “big K , little k ” structure of section 7.3. The individual state is employment status and the wage; the aggregate state is U , with perceived law $U' = G(U)$. The model has no quit option, so an employed worker keeps his wage until fired. (a) Let $V(U)$ be the value of an unemployed worker at the start of a period, before the draw, when unemployment is U . Let $E(w, U)$ be the value of working this period at wage w . A worker who works this period is fired at the start of next period with

probability λ ; he then draws immediately, so his value is $V(U')$. Hence

$$E(w, U) = w + \beta[(1 - \lambda)E(w, G(U)) + \lambda V(G(U))],$$

$$\boxed{V(U) = \int_0^B \max \left\{ E(w, U), c(U) + \beta V(G(U)) \right\} dF(w).}$$

The max compares accepting w with rejecting it: collect $c(U)$ and search again next period.

(b) Guess $E(w, U) = w/\kappa + e(U)$ with $\kappa \equiv 1 - \beta(1 - \lambda)$. Substituting,

$$\frac{w}{\kappa} + e(U) = w + \beta(1 - \lambda)\frac{w}{\kappa} + \beta[(1 - \lambda)e(G(U)) + \lambda V(G(U))].$$

The w terms match because $\frac{w}{\kappa}(1 - \beta(1 - \lambda)) = w$. The rest gives $e(U) = \beta[(1 - \lambda)e(G(U)) + \lambda V(G(U))]$. So E is strictly increasing in w with slope $1/\kappa$, while the value of rejecting does not depend on w . The optimal rule is therefore a **reservation wage** that depends on the aggregate state: accept iff $w \geq \bar{w}(U)$, where $E(\bar{w}(U), U) = c(U) + \beta V(G(U))$:

$$\boxed{\bar{w}(U) = \kappa[c(U) + \beta V(G(U)) - e(U)]}$$

(truncated to $[0, B]$). Substituting $\max\{E, c + \beta V'\} = c + \beta V' + \max\{w - \bar{w}, 0\}/\kappa$ into (a) gives $V(U) = c(U) + \beta V(G(U)) + \kappa^{-1} \int_{\bar{w}(U)}^B (w - \bar{w}(U)) dF(w)$. $\bar{w}$ depends on U through today's benefit $c(U)$ and, through G , on forecasts of future benefits and of future job values.

(c) All unemployed workers face the same U and use the same $\bar{w}(U)$. Their offers are i.i.d., so by the law of large numbers over the continuum the fraction who accept is

$$\boxed{\phi(U) = \text{Prob}(w \geq \bar{w}(U)) = 1 - F(\bar{w}(U)),}$$

and the actual law of motion is $U^* = G_A(U) \equiv \lambda(1 - U) + F(\bar{w}(U))U$. Because $\bar{w}$ depends on the perceived law G , so do ϕ and G_A .

(d) A **recursive competitive equilibrium** is a pair of value functions $V(U)$ and $E(w, U)$, a decision rule $\bar{w}(U)$, and a perceived law of motion $G(U)$ such that: (i) given G , V and E satisfy the Bellman equations in (a) and $\bar{w}(U)$ is the optimal reservation wage from (b); and (ii) the actual law implied by the rule equals the perceived one, $\lambda(1 - U) + F(\bar{w}(U))U = G(U)$ for all $U \in [0, 1]$. In the notation of section 7.3, the worker's state is $x = (\text{status}, w)$, the aggregate state is $X = U$, h is the rule "accept iff $w \geq \bar{w}(U)$ ", and $G_A = G$ is the fixed-point condition.

Steady state. If $G(U_s) = U_s$, then V, e are constants. From (b), $e = \beta\lambda V/\kappa$, so $\bar{w} = \kappa(c + \beta V) - \beta\lambda V = \kappa c + \beta(1 - \beta)(1 - \lambda)V$, using $\kappa - \lambda = (1 - \beta)(1 - \lambda)$. From (a), $(1 - \beta)V = c + \kappa^{-1} \int_{\bar{w}}^B (w - \bar{w}) dF$. Eliminating V :

$$\bar{w} = \kappa c + \beta(1 - \lambda) \left[c + \frac{1}{\kappa} \int_{\bar{w}}^B (w - \bar{w}) dF(w) \right], \quad c = c(U_s), \quad U_s = \frac{\lambda}{\lambda + \phi(U_s)},$$

where the last condition solves $U = \lambda(1 - U) + (1 - \phi)U$.

Worked example (parameters chosen: $\beta = .95$, $\lambda = .05$, F uniform on $[0, 1]$, so $\int_{\bar{w}}^1 (w - \bar{w})dw = (1 - \bar{w})^2/2$, and $c = 2$). The RCE is computed on a 1001-point grid for U . The inner loop solves the worker's Bellman equations for a given G (a contraction). The outer loop sets $G \leftarrow \frac{1}{2}G + \frac{1}{2}G_A$ and converges in 68 rounds. The equilibrium $\bar{w}(U)$ rises from 0.666 at $U = 0$ to 0.987 at $U = 1$, and G is increasing. The unique steady state is $U_s = 0.16705$, with $\bar{w}_s = 0.75069$, $\phi_s = 0.24931$ and $c(U_s) = 0.46302$. The slope of G there is 0.784, so unemployment converges monotonically but slowly. Benefits that rise with U make workers pickier when unemployment is high, which makes high unemployment more persistent.

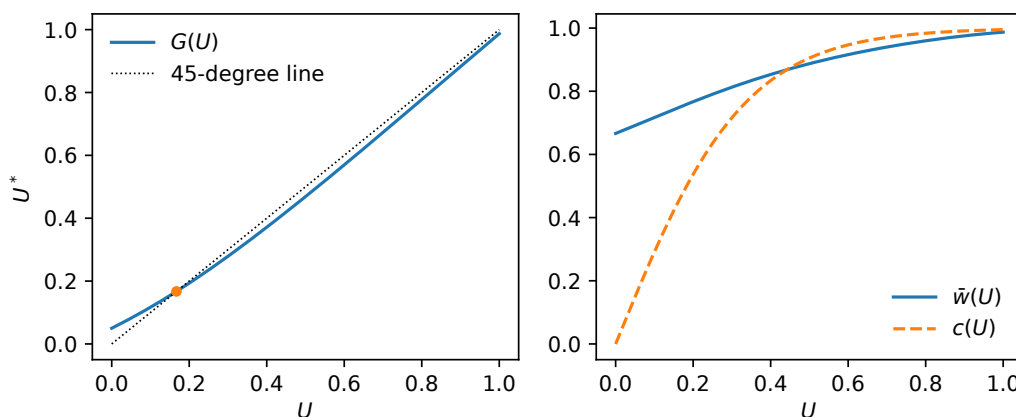


Figure 4: Exercise 7.7. Left: the equilibrium law of motion $U^* = G(U)$. Right: the reservation wage and the benefit schedule.

Reading of the timing. Under the stated timing, a worker hired this period is exposed to firing at the start of the next one. Exact accounting would then give $U^* = \lambda(1 - U + \phi U) + (1 - \phi)U$, but the book's law omits the term $\lambda\phi(U)U$. We take the stated law as the definition of the aggregate dynamics, as the exercise directs. The individual Bellman equations are unaffected. Under the alternative law the example's steady state would be $U_s = 0.1796$ instead of 0.1671.

Numerical exercise — see the companion code

Independent route: the scalar steady-state equations (Brent's method) give U_s and $\bar{w}_s$ equal to the grid RCE to 10^{-4} . A sign-change count on $(0, 1)$ confirms uniqueness, and a simulated path of U from 0.4 converges to U_s .

Companion code

Every numerical result above is reproduced by the notebook `ls_ch07.ipynb` in this folder (executed, outputs saved). Each claim is checked by an independent route with an `assert`.

Section	What it checks
Shared	<code>olrp</code> , <code>dare</code> , <code>evaluate</code> , <code>howard</code> copied from <code>ls_ch05.ipynb</code> ; new <code>nnash</code> (stacked Riccati).
7.1	Return mapping; <code>olrp</code> vs DARE vs undetermined coefficients; Euler residual; actual law for n firms.
7.2	$\mathcal{M}(H)$ at pairs (i)–(iii); quadratic fixed point; Jacobian by finite differences; damped vs undamped algorithm; figure.
7.3–7.4	Surplus by quadrature vs quadratic form; <code>olrp</code> vs DARE vs Euler closed forms; REE = planner.
7.5	<code>nnash</code> vs best-response iteration; DARE given the rival's rule; symmetry; steady state; figure.
7.6	Grid backward induction (MPE and commitment) vs log/Cobb–Douglas closed form; welfare ranking; figure.
7.7	Grid RCE vs scalar steady state; uniqueness; monotone $\bar{w}$, G ; alternative timing; figure.

Figures are saved to `fig/ls07_*.pdf` with TrueType (not Type 3) fonts.